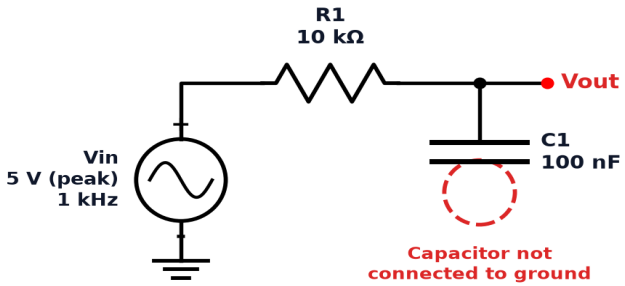
1. Overview

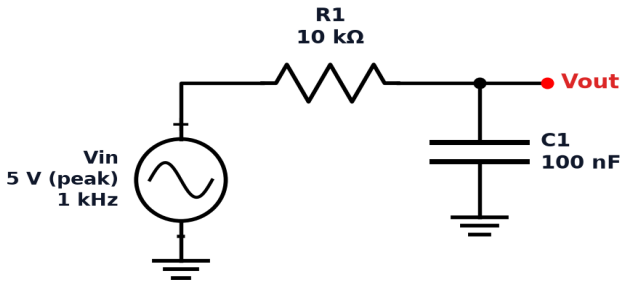
Circuit Type:	RC Low-Pass Filter	**Input File:**	rc_filter_open_cap.png
Analysis Status:	⚠ 1 Potential Issue Found	**Source:**	Image Upload

2. Circuit Comparison

Uploaded Circuit (Input)



Expected / Reference Circuit



3. Detected Issues

#	Severity	Component / Location	Finding	Recommended Action
1	Error	C1 (lower terminal)	Capacitor is not connected to ground (open circuit).	Connect the lower terminal of C1 to ground as shown in the reference circuit.

4. Extracted Components

Reference	Component Type	Extracted Value	Unit	Status	Notes
R1	Resistor	10	kΩ	Verified	Value clearly identified
C1	Capacitor	100	nF	Verified	Value clearly identified
Vin	AC Source (Sine)	5 (peak), 1 kHz	V, Hz	Verified	Input source identified

5. Expected Behaviour (Theory)

For an rc low-pass filter, the cutoff frequency is given by:

$$f_c = \frac{1}{2\pi RC}$$

Using extracted values:

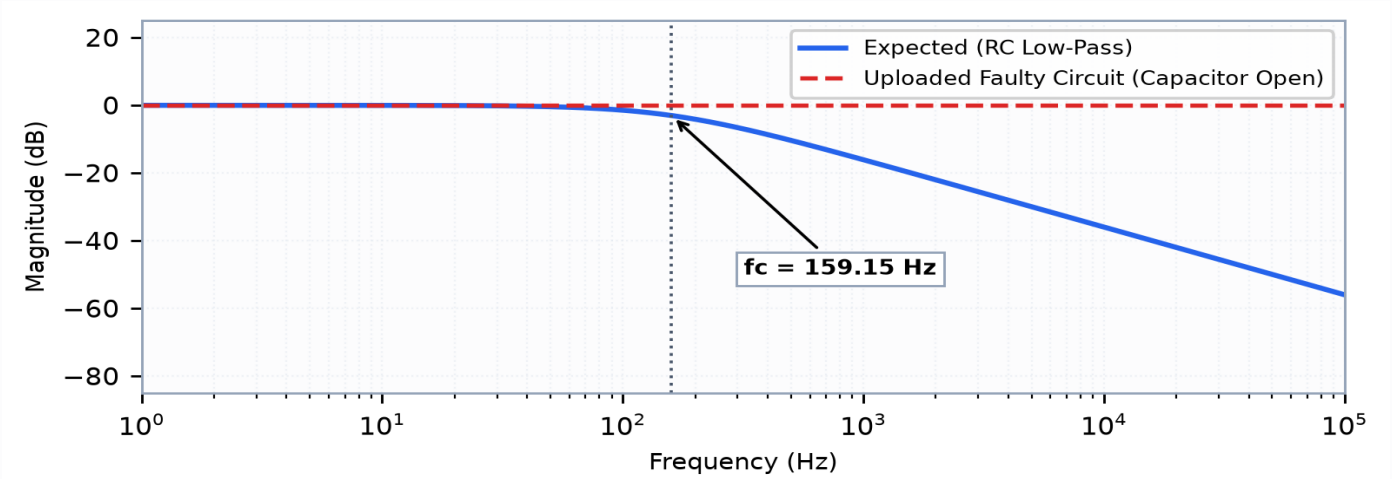
$R = 10\ \text{k}\Omega = 10,000\ \Omega$

$C = 100\ \text{nF} = 100 \times 10^{-9}\ \text{F}$

$f_c = \frac{1}{2\pi(10,000)(100 \times 10^{-9})} \approx 159.15\ \text{Hz}$

6. Simulation Results

Frequency Response (Magnitude)



7. Recommended Actions

1. Connect the lower terminal of C1 to ground.
2. Verify component values: R1 = 10 kΩ, C1 = 100 nF.
3. Re-run simulation or measure the output to confirm low-pass behaviour.
4. Ensure all junctions are properly connected in the physical circuit (no open connections).

8. Conclusion

The uploaded schematic represents an intended RC low-pass filter, but the capacitor is not connected to ground. Due to this open connection, the circuit will not exhibit the expected low-pass behaviour. Connecting C1 to ground will resolve the issue.